

MADHUMITHA SIVANATHAN

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SUMMARY

B.Tech Information Technology student building and deploying production-relevant machine learning systems, currently developing an LSTM-based time-series forecasting system to predict next-day solar energy production for a PV plant as a Student Intern. Experienced in real-time computer vision through RTSP-based fire detection and a YOLOv8-based medical imaging project achieving 94%+ detection accuracy, with hands-on experience across data preprocessing, feature engineering, model development, inference, and evaluation. Comfortable with backend and API development using Django, Express.js, and Next.js.

EXPERIENCE

Student Intern | Aattral Research Private Limited July 2026 – Present

- Engineered a real-time fire detection pipeline for Energy Management System (EMS) applications using RTSP-based IP camera streams, integrating deep learning inference for continuous industrial safety monitoring.
- Architected an RTSP stream simulator using MediaMTX and FFmpeg to emulate multiple IP camera feeds, enabling scalable testing, debugging, and validation of real-time video analytics pipelines without physical camera hardware.
- Currently developing an LSTM-based time-series forecasting system to predict next-day solar energy production at the inverter level, incorporating historical weather, inverter telemetry, and forecast data for PV plant energy forecasting.

Research Intern | Cantier Systems Private Limited May 15 – May 30, 2026

- Evaluated software architecture and MySQL database designs to support development of a KPI Management Platform built with Next.js and Express.js.
- Developed structured REST API workflows and integrated frontend and backend components to enable efficient real-time data exchange.
- Applied AI product development principles to design and refine application features based on real-time operational requirements.

PROJECTS

Automated Brain Tumor Diagnosis using YOLO-based Deep Learning Model | [GitHub](#)

- Built a YOLOv8-based object detection system to identify and localize brain tumors from MRI images, from data preparation through model evaluation.
- Curated and annotated a 4,737-image training set and 512-image test set in YOLO format using Roboflow.
- Achieved 94.66% precision, 94.44% recall, and 96.41% mAP@0.5 (80.47% mAP@0.5:0.95), demonstrating high detection accuracy and robustness.
- Delivered bounding-box-level tumor localization to support faster, more reliable diagnostic workflows.

Tech Stack: Python, TensorFlow, YOLOv8, Computer Vision | Tools: VS Code, Roboflow

RTSP-Based Real-Time Fire Detection

- Built a real-time fire detection pipeline for 3 simulated IP camera feeds using the RTSP protocol, enabling continuous video monitoring and automated fire and smoke detection.
- Designed an RTSP stream simulator using MediaMTX and FFmpeg to emulate multiple concurrent camera feeds, enabling scalable testing without physical camera hardware.
- Integrated a pretrained YOLO-based model for 2-class fire and smoke detection, processing live video frames through the RTSP pipeline for real-time inference.

Tech Stack: Python, YOLO, OpenCV, RTSP | Tools: MediaMTX, FFmpeg, Django, VS Code

SKILLS

Programming Languages: Java, C, Python

Machine Learning & Computer Vision: YOLOv8, TensorFlow, Object Detection, Image Processing, Real-Time Video Analytics

Frameworks & Libraries: React, React Native, Next.js, Node.js, Express.js, Django

Web Technologies: HTML, CSS

Tools & Platforms: VS Code, Git (GitHub, GitLab), Firebase, Roboflow, Android Studio, REST APIs

Core Concepts: Data Structures, Problem Solving

EDUCATION

Dr. N.G.P. Institute of Technology 2024 – 2028

Bachelor of Technology in Information Technology — Coimbatore, India

APS Academy Matriculation Higher Secondary School 2021 – 2024

HSC & SSLC — Tirupur, India

CERTIFICATIONS

- Programming in Java — NPTEL (IIT Kharagpur), Elite + Gold
- Problem Solving Through Programming in C — NPTEL (IIT Kharagpur)
- Generative AI and Python Programming — Navigate Labs (Value-Added Course)

ACHIEVEMENTS

- Awarded Third Prize at an intercollegiate tech hackathon for the project Farm Hands.
- Achieved Second Place in the SpectraX Project Expo conducted by KPR Institute of Engineering and Technology (KPRIET).